Forecasting Stock Market Returns: The Sum of the Parts is More Than the Whole

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Research Motivation

- **Background:** Many studies attempt to forecast stock market returns using predictive regressions with variables such as dividend yields or earnings-price ratios.
- **Problem:** Out-of-sample performance of these models is often poor. Goyal and Welch (2008) found that the historical mean frequently outperforms predictive regressions.
- **Objective:** This paper introduces the **Sum-of-the-Parts (SOP) method**, which forecasts the components of stock returns separately to improve predictive power.

Research Question

- **Main Question:** Can forecasting the components of stock returns separately improve out-of-sample predictability?
- **Motivation:** Traditional predictive regressions often fail due to high persistence in variables like the dividend–price ratio and estimation errors.
- **Proposed Solution:** The SOP (Sum-of-the-Parts) method decomposes total returns into predictable elements — dividend yield, earnings growth, and price–earnings growth — and forecasts them individually.
- **Expected Outcome:** Improved forecasting accuracy, reduced estimation error, and more stable performance across time.

Methodology Overview

- **Concept:** Decompose total stock market return into three additive components:

$$r_{t+1} = gm_{t+1} + ge_{t+1} + dp_{t+1}$$

  where:
  - gm_{t+1} = price–earnings ratio growth
  - ge_{t+1} = earnings growth
  - dp_{t+1} = dividend–price ratio

- **Forecasting logic:**

$$\hat{m}_s = \hat{ge}_s + \hat{dp}_s$$

  assuming no multiple growth ($\hat{gm}_s = 0$).

- **Intuition:** Instead of forecasting total returns directly, SOP predicts each underlying driver (dividend yield and earnings growth) separately and then aggregates their forecasts.

- **Idea:** Each component has different persistence, allowing tailored forecasting methods to reduce estimation error.

The Coefficient Puzzle: β (3.747 vs 1)

- **Traditional Predictive Regression:**

$$r_{t+1} = \alpha + \beta dp_t + \varepsilon_{t+1}$$

Empirical estimates (1927–2007): $\hat{\beta} = 3.747$ — far above the theoretical value of 1.

- **Theoretical Expectation:** From the log-linear return identity (Campbell–Shiller):

$$r_{t+1} \approx \rho dp_{t+1} - dp_t + \Delta d_{t+1}$$

The coefficient on dp_t should be **1**, not 3.747.

- **Implication:** Traditional regressions overstate predictability — β is inflated by persistence and noise in dp_t.

The Coefficient Puzzle: α and SOP Consistency

- **SOP Framework:**

$$r_{t+1} = \log(1 + R_{t+1}) = gm_{t+1} + ge_{t+1} + dp_{t+1}$$

— no need to estimate α or β directly; return components mechanically sum to the total return.

- **Comparison:**
  - Traditional regression: α : -0.018 , $\beta \approx 3.7$ (overstated sensitivity).
  - SOP: α : 0.062 , $\beta \approx 1$ (theoretically consistent).

- **Implication:** SOP aligns both α and β with the log-linear return identity, reducing estimation bias and improving interpretability.

Data and Sample

- **Data Source:** Goyal and Welch (2008) dataset, widely used in return predictability literature.
- **Sample Period:** 1927–2007.
- **Frequency:** Monthly and annual (non-overlapping) observations.
- **Market Proxy:** S&P 500 total return index (including dividends).
- **Key Variables:**
  - Dividend–price ratio (dp)
  - Earnings growth (ge)
  - Price–earnings ratio growth (gm)
- **Descriptive Findings:** Dividend–price ratio is highly persistent; earnings and multiple growth are less persistent.

Empirical Results (Monthly)

- **Out-of-sample performance:** SOP method achieves an R^2_{OOS} of **1.32%** using monthly data.
- **Benchmark comparison:** Predictive regressions show negative or near-zero out-of-sample R^2 (ranging from -1.78% to 0.69%).
- **Shrinkage adjustment:** Improves regressions slightly, but still underperforms SOP.
- **Interpretation:** The SOP's advantage arises from avoiding parameter estimation errors — forecasts rely on observed ratios and long-run averages instead of estimated coefficients.
- **Takeaway:** Even with short-horizon (monthly) data, the SOP method produces consistent and positive predictive power.

Empirical Results (Annual)

- **Out-of-sample performance:** The SOP method achieves an impressive annual R^2_{OOS} of **13.43%**.
- **Predictive regressions:** Traditional models produce mostly negative R^2 values (from -17.57% to 7.54%), confirming poor out-of-sample predictability.
- **Component contributions:**
  - Dividend–price ratio contributes about 43% of forecast variance.
  - Earnings growth contributes about 56%.
  - Covariance between components is minimal (1%).
- **Interpretation:** Both components play roughly equal roles in driving forecast improvements.
- **Key insight:** SOP significantly enhances long-term predictability with a simple, low-error framework.

Extensions

- **Incorporating Price–Earnings Growth (gm):** Two extensions to include multiple growth forecasts were tested:
  1. **Multiple Growth Regression:** Forecast gm using predictive regression techniques.
  2. **Multiple Reversion:** Assume price–earnings ratio reverts toward its expected level given macroeconomic conditions.
- **Results:**
  - SOP with multiple growth regression: R^2_{OOS} up to 14.4%.
  - SOP with multiple reversion: R^2_{OOS} up to 16.9%.
- **Interpretation:** Including PE ratio dynamics improves long-horizon forecasts, especially when accounting for mean reversion.
- **Conclusion:** SOP extensions remain robust and economically significant.

Economic Implications

- **Practical relevance:** The SOP method's accurate forecasts are useful for portfolio allocation and cost-of-capital estimation.
- **Investment Performance:**
  - SOP-based market timing strategy improves the Sharpe ratio by approximately 0.3.
  - Certainty equivalent gain of about 1.8% per year relative to the historical mean strategy.
- **Interpretation:** Forecasting return components separately yields tangible economic benefits.
- **Broader Impact:** Demonstrates that simple, low-error models can enhance real-world financial decision-making.

Sharpe Ratio Gains and Trading Performance

Table 6
Trading strategies: Sharpe ratio gains.
This table presents out-of-sample portfolio choice results at monthly and annual (nonoverlapping) frequencies from predictive regressions and the sum-of-the-parts (SOP) method. The numbers are the change in Sharpe ratio of a trading strategy timing the market with different return forecasts relative to timing the market with the historical mean return. All numbers are annualized. The sample period is from December 1927 through December 2007. Forecasts begin 20 years after the sample start.

Variable	Description	Predictive regression	Predictive regression (shrinkage)	SOP no multiple growth	SOP multiple growth regression	SOP multiple reversion
Panel A: Monthly return forecasts (January 1948–December 2007)						
	–	–	–	0.31	–	–
SVAR	Stock variance	0.00	0.00	–	0.12	0.22
DFR	Default return spread	−0.06	−0.01	–	0.30	0.24
LTY	Long-term bond yield	−0.25	−0.06	–	0.29	0.09
LTR	Long-term bond return	−0.12	−0.02	–	0.23	0.24
INFL	Inflation	−0.04	−0.01	–	0.31	0.19
TMS	Term spread	−0.05	−0.02	–	0.28	0.23
TBL	T-bill rate	−0.18	−0.04	–	0.32	0.16
DFY	Default yield spread	−0.02	0.00	–	0.33	0.24
NTIS	Net equity expansion	0.04	0.06	–	0.28	0.24
ROE	Return on equity	−0.06	−0.02	–	0.27	0.12
DE	Dividend payout	−0.02	0.00	–	0.32	0.14
EP	Earnings price	−0.09	0.30	–	0.23	–
SEP	Smooth earnings price	−0.21	0.12	–	0.12	–
DP	Dividend price	0.11	0.08	–	0.14	–
DY	Dividend yield	−0.13	0.15	–	0.07	–
BM	Book-to-market	−0.34	0.04	–	0.01	–
	Constant	–	–	–	–	0.24
Panel B: Annual return forecasts (1948–2007)						
	–	–	–	0.22	–	–
SVAR	Stock variance	0.01	0.00	–	0.23	0.11
DFR	Default return spread	0.03	0.02	–	0.23	0.12
LTY	Long-term bond yield	−0.14	−0.03	–	0.19	0.02
LTR	Long-term bond return	0.08	0.06	–	0.23	0.15
INFL	Inflation	0.01	0.02	–	0.21	0.09
TMS	Term spread	−0.10	−0.02	–	0.18	0.15
TBL	T-bill rate	−0.19	−0.04	–	0.19	0.08
DFY	Default yield spread	−0.01	−0.01	–	0.24	0.13
NTIS	Net equity expansion	0.05	0.04	–	0.22	0.12
ROE	Return on equity	−0.15	−0.04	–	0.16	0.03
DE	Dividend payout	0.00	0.00	–	0.21	0.04
EP	Earnings price	0.05	0.15	–	0.12	–
SEP	Smooth earnings price	−0.15	0.07	–	0.06	–
DP	Dividend price	−0.02	0.02	–	0.04	–
DY	Dividend yield	−0.21	0.07	–	0.20	–
BM	Book-to-market	−0.19	0.03	–	0.09	–
	Constant	–	–	–	–	0.13

Time Stability of Predictability

- **Issue:** Traditional predictive models often lose forecasting power in recent decades (Goyal and Welch, 2008).
- **Approach:** Divide the sample into early (1948–1976) and late (1977–2007) periods to test stability.
- **Findings:**
  - SOP maintains positive out-of-sample R^2 in both subperiods.
  - Predictive regressions mostly fail in the later decades.
- **Interpretation:** Predictive power of SOP is not driven by early-sample overfitting.
- **Conclusion:** SOP remains economically meaningful and statistically robust across time.
- **Implication:** This finding suggests that SOP captures structural relationships in returns rather than sample-specific artifacts.

Annual Return Forecasts: Early Subsample (1948–1976)

Variable	Description	In-sample R-square	Out-of-sample R-square: Predictive regression	Predictive regression (shrinkage)	SOP no multiple growth	SOP multiple growth regression	SOP multiple reversion
Panel B: Annual return forecasts (1948–1976)							
	–	–	–	–	14.66***	–	–
SVAR	Stock variance	0.19	−0.76	−0.21	–	13.93***	21.54***
DFR	Default return spread	2.34	4.52*	1.66	–	14.82***	20.17***
LTY	Long-term bond yield	0.70	−10.95	−0.82	–	9.62**	13.40***
LTR	Long-term bond return	6.82*	9.64*	5.10**	–	13.44***	25.18***
INFL	Inflation	1.49	−0.99	0.72	–	13.77***	22.18***
TMS	Term spread	1.91	−6.66	−0.68	–	12.80***	21.85***
TBL	T-bill rate	1.59	−12.14	−1.43	–	11.66**	18.31***
DFY	Default yield spread	0.05	−1.64	−0.43	–	14.56***	21.98***
NTIS	Net equity expansion	14.91***	0.65	0.31	–	14.59***	21.75***
ROE	Return on equity	0.91	−12.62	−1.93	–	14.73***	22.82***
DE	Dividend payout	1.30	−0.23	−0.12	–	13.09***	21.52***
EP	Earnings price	6.74*	14.14***	4.71*	–	21.77***	–
SEP	Smooth earnings price	22.44***	−10.42	5.91**	–	23.21***	–
DP	Dividend price	2.93	4.48*	1.82	–	21.02***	–
DY	Dividend yield	5.28	−17.74	4.34*	–	18.16***	–
BM	Book-to-market	14.73***	8.30***	4.41*	–	19.87***	–
	Constant	–	–	–	–	–	21.76***

Annual Return Forecasts: Later Subsample (1977–2007)

Variable	Description	In-sample R-square	Out-of-sample R-square: Predictive regression	Predictive regression (shrinkage)	SOP no multiple growth	SOP multiple growth regression	SOP multiple reversion
Panel D: Annual return forecast (1977–2007)							
	–	–	–	–	12.10***	–	–
SVAR	Stock variance	0.71	−25.88	−1.32	–	10.83**	7.28**
DFR	Default return spread	3.15	−5.65	−0.58	–	13.56***	11.51**
LTY	Long-term bond yield	2.37	−3.39	−0.09	–	11.55***	7.37**
LTR	Long-term bond return	3.26	−21.50	−0.15	–	12.35***	10.09**
INFL	Inflation	2.24	−10.39	−0.80	–	9.64**	11.64***
TMS	Term spread	1.27	−15.70	−2.04	–	9.92**	10.75**
TBL	T-bill rate	0.51	−17.57	−2.53	–	9.24**	8.81**
DFY	Default yield spread	5.71*	−14.77	−0.22	–	8.83**	9.38**
NTIS	Net equity expansion	2.53	1.53	3.30*	–	10.76**	8.08**
ROE	Return on equity	0.45	−9.68	0.43	–	15.32***	5.13**
DE	Dividend payout	0.14	−9.82	−1.79	–	11.35***	−7.56
EP	Earnings price	8.42**	1.82	3.49*	–	9.83**	–
SEP	Smooth earnings price	7.11*	−12.50	1.39	–	5.85**	–
DP	Dividend price	6.97*	−26.89	0.62	–	0.01	–
DY	Dividend yield	5.69*	−15.74	0.52	–	6.60**	–
BM	Book-to-market	3.04	−11.16	−0.50	–	6.33**	–
	Constant	–	–	–	–	–	9.74**

Robustness and Simulation Evidence

- **Robustness Checks:**
  - SOP consistently outperforms predictive regressions under different forecasting horizons (monthly, annual).
  - Results hold for various moving-average windows (10–20 years).
  - Remain stable under alternative autoregressive assumptions for dp_t.
- **Simulation Results:**
  - Monte Carlo simulations show that SOP forecasts have lower mean squared error (MSE).
  - SOP's prediction intervals are tighter and closer to realized returns.
- **Interpretation:** Across tests of time horizon, estimation method, and simulated data, **SOP consistently delivers higher predictive accuracy and lower estimation error.**

Cross-Market and Analyst Forecast Robustness

- **International Evidence:**
  - SOP tested on major international markets — United States, United Kingdom, and Japan.
  - Out-of-sample R^2 remains positive and significant across all three countries.
- **Analyst Forecast Comparison:**
  - Analysts' return forecasts tend to be **systematically optimistic**.
  - SOP forecasts are unbiased and exhibit smaller forecast errors.
- **Conclusion: Across time, methods, and markets,** SOP demonstrates robust and generalizable forecasting power.

International Evidence: Forecasting Stock Market Returns

Table 7
Forecasts of stock market returns: international evidence.

This table presents in-sample and out-of-sample *R*-squares (in percentage) for stock market return forecasts at annual (nonoverlapping) frequency in the UK (Panel A), Japan (Panel B), and the US (Panel C) from predictive regressions and the sum-of-the-parts (SOP) method. The in-sample *R*-squares are estimated over the full-sample period. The out-of-sample *R*-squares compare the forecast error of the model with the forecast error of the historical mean. The sample period is from 1950 or 1960 through 2007. Forecasts begin 20 years after the sample start. Asterisks denote significance of the in-sample regression as measured by the *F*-statistic or significance of the out-of-sample *MSE–F* statistic of McCracken (2007). ***, **, and * denote significance at the 1%, 5%, and 10% level respectively.

				Out-of-sample *R*-square				
Variable	Description	Forecast start	In-sample *R*-square	Predictive regression	Predictive regression (shrinkage)	SOP no multiple growth	SOP multiple growth regression	SOP multiple reversion
Panel A: UK annual return forecasts								
	–	1970	–	–	–	10.73***	–	–
LTY	Long-term bond yield	1970	5.29*	−47.54	−5.61	–	4.16**	11.27***
TMS	Term spread	1970	3.10	−14.71	−1.13	–	9.26**	11.60***
TBL	T-bill rate	1970	1.47	−20.87	−3.07	–	6.39**	11.51***
DY	Dividend yield	1970	11.97***	−9.19	5.07**	–	13.28***	10.78***
	Constant	1970	–	–	–	–	–	11.75***
Panel B: Japan annual return forecasts								
	–	1970	–	–	–	12.14***	–	–
LTY	Long-term bond yield	1970	1.69	−11.01	−1.86	–	12.11***	11.87***
TMS	Term spread	1980	0.36	−5.46	−0.89	–	5.75**	5.82**
TBL	T-bill rate	1980	1.76	−7.57	−0.62	–	5.14*	5.62**
DY	Dividend yield	1970	15.24***	3.12*	6.63**	–	10.25***	11.99***
	Constant	1970	–	–	–	–	–	11.91***
Panel C: US annual return forecasts								
	–	1970	–	–	–	7.75**	–	–
LTY	Long-term bond yield	1970	0.17	−20.73	−1.51	–	4.47**	3.12*
TMS	Term spread	1970	1.11	−12.05	−0.99	–	8.24**	5.50**
TBL	T-bill rate	1970	0.03	−21.18	−2.00	–	5.06**	3.40*
DY	Dividend yield	1970	7.95**	0.96	2.68*	–	6.64**	5.73**
	Constant	1970	–	–	–	–	–	5.92**

Overall Contribution

- **Methodological:** Introduces a simple yet effective decomposition-based forecasting approach.
- **Empirical:** Demonstrates strong and robust out-of-sample predictability across multiple decades and market regimes.
- **Practical:** Provides a transparent framework for investors to separate and understand sources of return predictability.

Reference

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Key Figures Summary

- **Monthly R^2_{OOS}:** 1.32%
- **Annual R^2_{OOS}:** 13.43%
- **Sharpe Ratio Gain:** +0.3
- **Certainty Equivalent Gain:** +1.8% per year
- **Main Contributors:** Dividend–price ratio (43%), earnings growth (56%)
- **Extensions:** SOP with multiple reversion achieves up to 16.9% R^2_{OOS}.